

Multiple carbapenemases in *Enterobacterales*

Nicole Lermaniaux^{1,2,3}, Ken Fakharuddin³, Erin McGill⁴, Yves Longtin⁵, Laura Mataseje³, the CNISP CPO working group, the NEKSUS consortium*, Katie L. Hopkins^{2,6}, Nicole Stoesser^{1,2}, Kevin Chau^{1,2}, Dorottya Nagy^{1,2}, Samuel Lipworth^{1,2}

¹Nuffield Department of Medicine, University of Oxford, Oxford, Oxfordshire, United Kingdom, ²NIHR Health Protection Research Unit in Healthcare Associated Infections and Antimicrobial Resistance, University of Oxford, Oxford, Oxfordshire, United Kingdom, ³National Microbiology Laboratory, Public Health Agency of Canada, Winnipeg, Manitoba, Canada, ⁴Public Health Agency of Canada, Ottawa, Ontario, Canada, ⁵Jewish General Hospital, Montréal, Québec, Canada, ⁶Antimicrobial Resistance and Healthcare Associated Infections Reference Unit, United Kingdom Health Security Agency, London, United Kingdom.

INTRODUCTION

- Carbapenemase genes confer resistance to last-resort treatment options for multi-drug-resistant *Enterobacterales* infections
- Major carbapenemase families (IMP, KPC, NDM, OXA, VIM) have varying hydrolytic profiles and inhibitor susceptibilities and therefore require different clinical treatment strategies¹
- The emergence of isolates encoding multiple carbapenemase genes is complicating treatment by further limiting effective antimicrobials², and these genomes have not yet been examined at scale
- We aimed to explore the genomic and plasmid context of global multiple carbapenemase-encoding *Enterobacterales*

METHODS

- Downloaded NCBI GenBank *Enterobacterales* genomes with ≤ 20 contigs (representing complete genomes, n= 46,517) and screened them for carbapenemase genes (IMP, KPC, NDM, OXA, VIM) with AMRFinderPlus³
- Kept genomes with two or more hits to the same type or different types of carbapenemases and included 23 genomes from the NEKSUS study
- Investigated plasmids with MOB-suite⁴ and pling⁵ and mobile elements with MobileElementFinder⁶ and IntegronFinder⁷

RESULTS

- We obtained 1,511 genomes (3.2 % of NCBI genomes screened) with two or more carbapenemases from 12 genera isolated between 2008 – 2025, representing 244 distinct allelic carbapenemase combinations (Figure 1)
- Of all plasmids in these genomes (n=8,086), 2,052 carried one carbapenemase, 424 carried two or more, and 5,610 carried none
- Most genomes carried carbapenemases on two plasmids (59.2 %, 896/1,511) or one plasmid (23.5 %, 354/1,511) (Figure 1)
- Plasmids with 2+ carbapenemase genes harboured significantly more antimicrobial resistance (AMR) genes and more AMR gene classes than plasmids with one or zero carbapenemases (median of 1 additional AMR gene class and 4 additional AMR genes)
- Multi-carbapenemases have diffused across many different plasmid structures (2,476 carbapenemase plasmids in 315 pling subcommunities) and 62.9 % (95/151) of non-singleton carbapenemase subcommunities also contain non-carbapenemase plasmids, including the largest subcommunity comprising 577 plasmids (Figure 3)
- The top 8 carbapenemase plasmid clusters comprised 37.5 % (928/2,476) of carbapenemase plasmids (Table 1), which are all known plasmid groups
- Carbapenemase genes were associated with diverse mobile elements, including integrons (IMP-4 and VIM-1), transposons (KPC-2, KPC-3, OXA-181, OXA-48), insertion sequences (OXA-232), or multiple different contexts (NDM-1 and NDM-5) (Figure 4)

CONCLUSIONS

- Multi-carbapenemase *Enterobacterales* are diverse, with respect to their allele combinations, mobile element associations, and plasmid backbones, and this may be facilitating their emergence and dissemination
- Single plasmids can carry multiple carbapenemase genes along with other AMR genes, which expands their resistance profiles and may provide a selective advantage
- Carbapenemase plasmids in these isolates often clustered with non-carbapenemase plasmids, indicating that there is a potential network for wider mobilisation of carbapenemases

Figure 1: Most common carbapenemase combinations coloured by genomic location. “Plas” = plasmid, “chr” = chromosome.

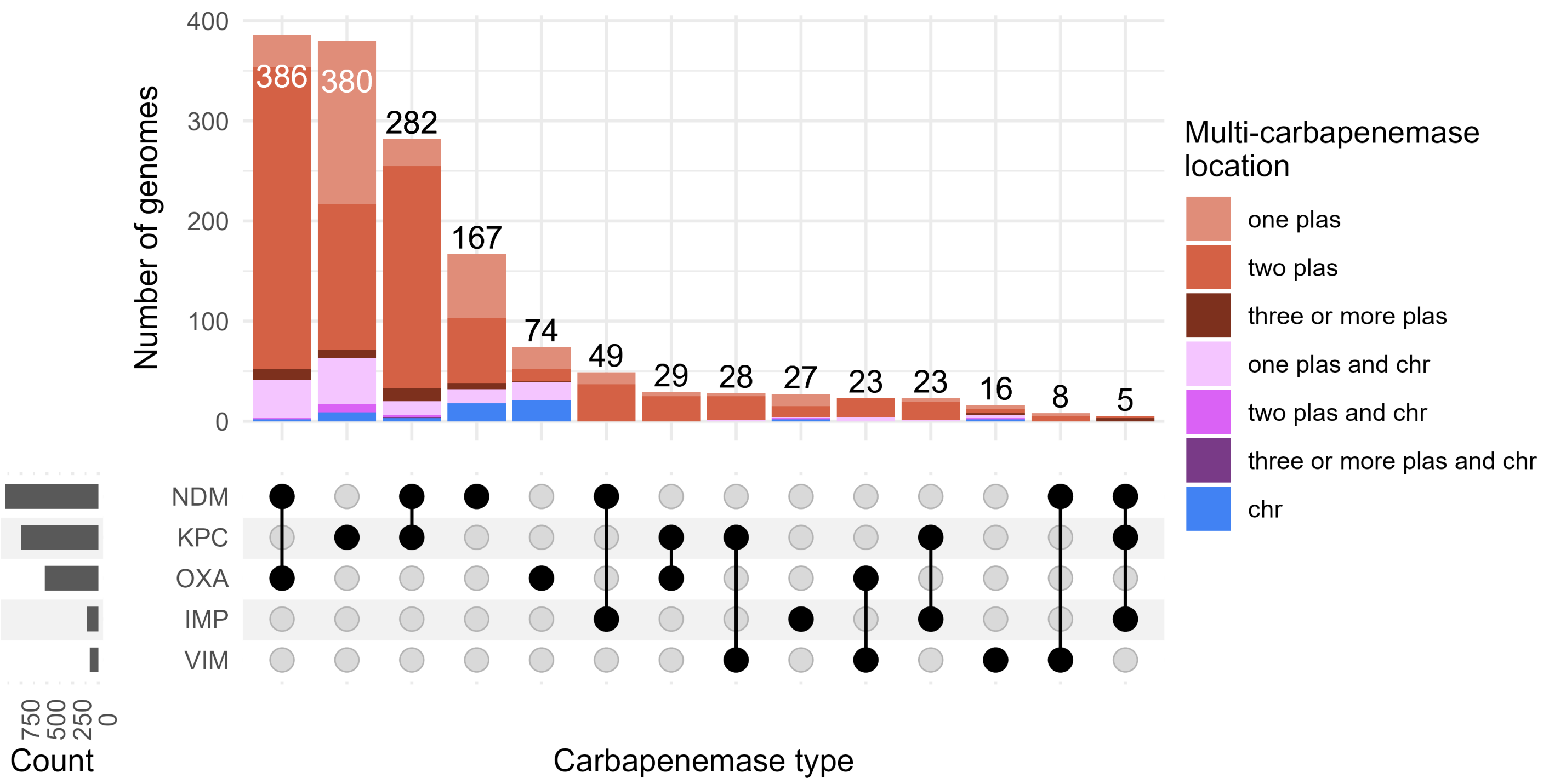


Figure 2: Network of the largest pling subcommunity (IncX3 replicons, median 51.5 kb size, 577 plasmids) containing plasmids with zero, one, or two or more carbapenemases (156, 343, and 58 plasmids, respectively) from multi-carbapenemase isolates.

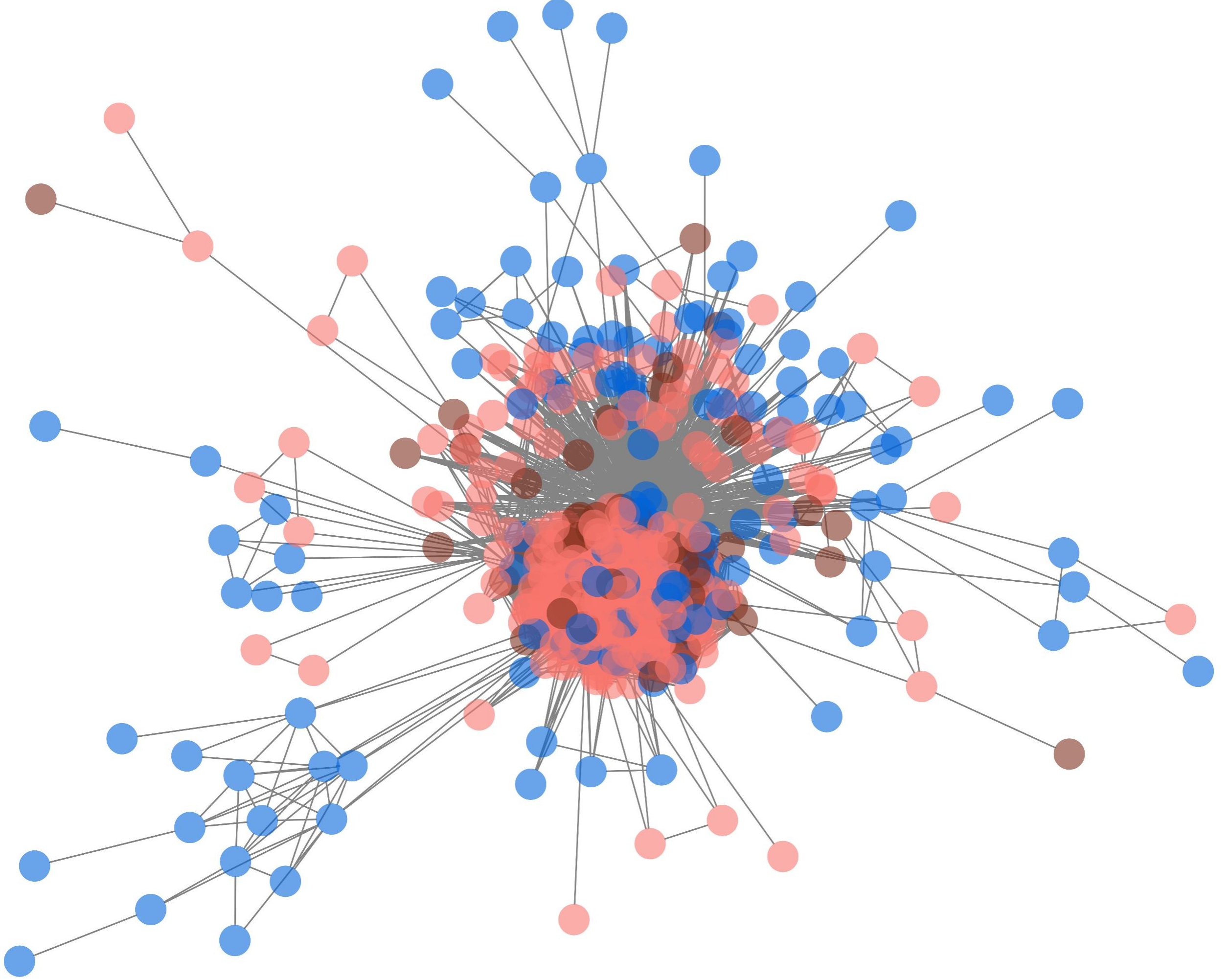
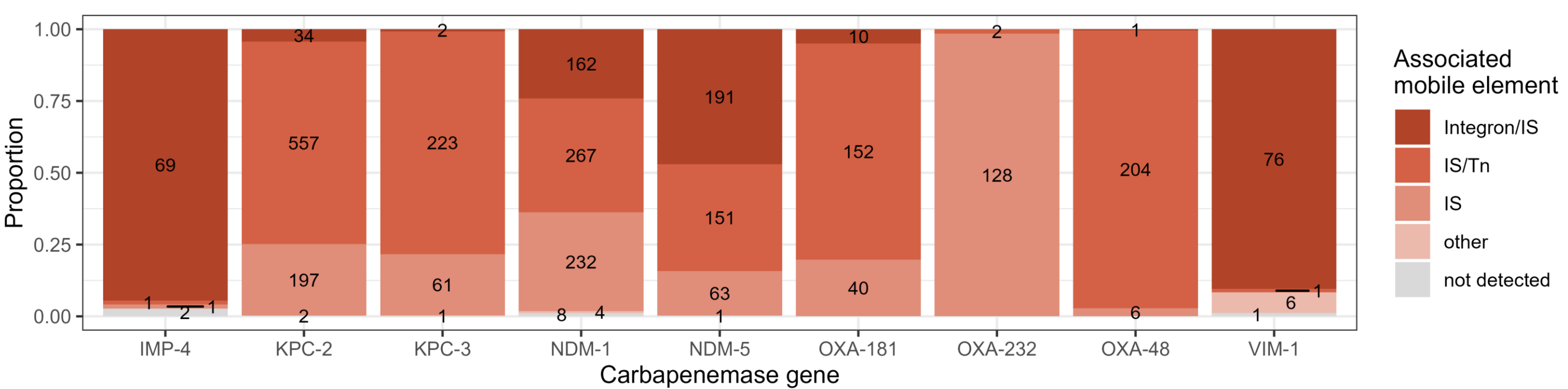


Table 1: Features of the top eight carbapenemase plasmid clusters in multi-carbapenemase genomes

MOB-suite secondary plasmid cluster	Number of plasmids	Most common replicon	Most common carbapenemase	Mean number of carbapenemases	Mean number of AMR genes	Number of genera	Number of years	Number of countries
AH615	228	IncX3/rep_cluster_1195	OXA-181	1.1	2.7	4	14	32
AH539	125	IncL/M	OXA-48	1.0	1.5	4	14	21
AL194	119	rep_cluster_1195	OXA-232	1.2	1.1	2	14	20
AH594	76	IncFIA/N/U	KPC-2	1.3	5.1	2	10	2
AJ272	66	IncC	NDM-1	1.2	8	6	12	15
AI436	60	IncFIB/HI1B/rep_cluster_1254	NDM-5	1.1	13.2	1	10	21
AH561	58	IncFIB/FII/rep_cluster_2183	KPC-2	1.3	2.8	3	14	12
AI283	58	IncFIA/FIC	NDM-5	1.0	8.8	3	13	17

Figure 3: Association between flanking regions of major carbapenemase genes and mobile elements (insertion sequence = IS, transposon = Tn)



References: ¹doi.org/10.1093/cid/ciad428; ²doi.org/10.1128/spectrum.00008-24; ³doi.org/10.1038/s41598-021-91456-0; ⁴doi.org/10.1099/mgen.0.000435; ⁵doi.org/10.1099/mgen.0.001300; ⁶doi.org/10.1093/jac/dkaa390; ⁷doi.org/10.3390/microorganisms10040700